Sheet Metal Engineering

ME 750: Course Project Presentation

Under the guidance of
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Design and Manufacturing of Vapor Chamber Thermal Plate



1. Objective

This study aims to achieve

Defect Prediction

- Identify forming defects (thinning, wrinkling, cracks) early
- Predict failure zones using FLD analysis
- Validate part formability before tool manufacturing

Process Optimization

- Tune binder force, drawbeads & blank shape
- Improve material flow and reduce springback
- Establish a robust, production-ready process window

Goal: Deliver a defect-free, formable part with a stable and repeatable process.

2. Input Setup – Material Model

Material data and hardening behaviour

Material: copper

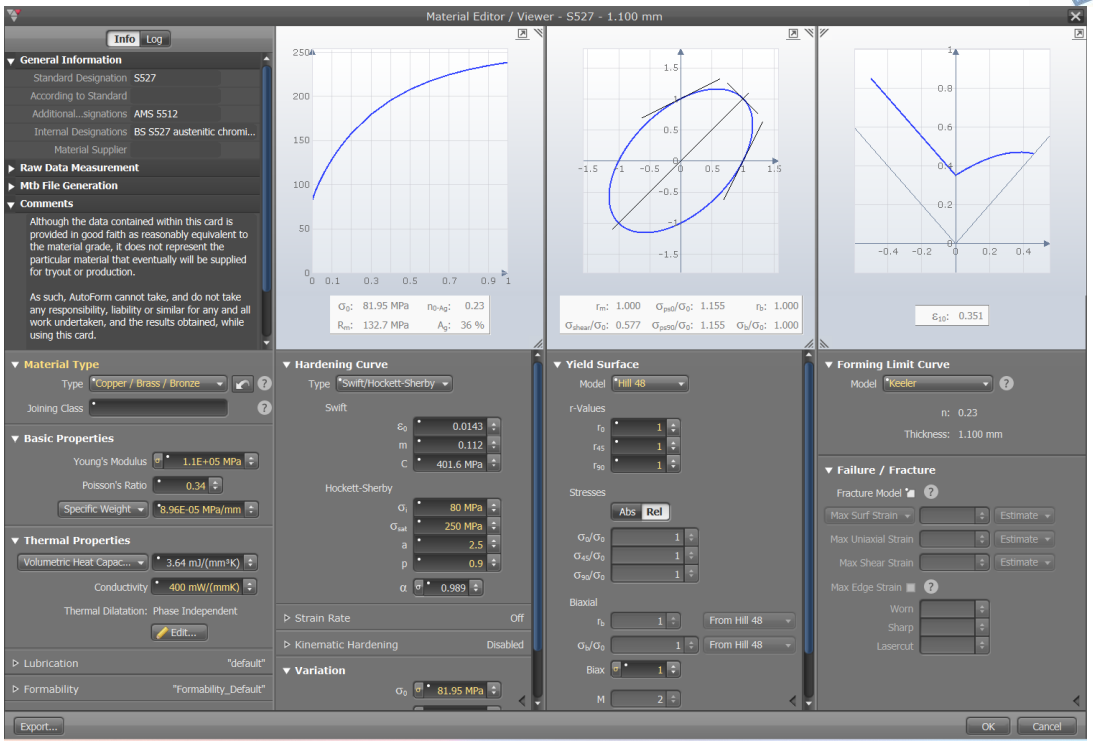
Yield Criterion: Hill ’48 (isotropic)

Hardening Law: $\sigma = C(\varepsilon_0 + \varepsilon_p)^m$

Key Properties:

E, ν, ρ, K, n, ε_0, r_0 / r_{45} / r_{90}

Source: Properties from the internet



3. Input Setup – Blank, Tools & Boundary Conditions

Geometric and contact inputs

Blank

Squared Shape, 160*160*1.2 mm

Tools

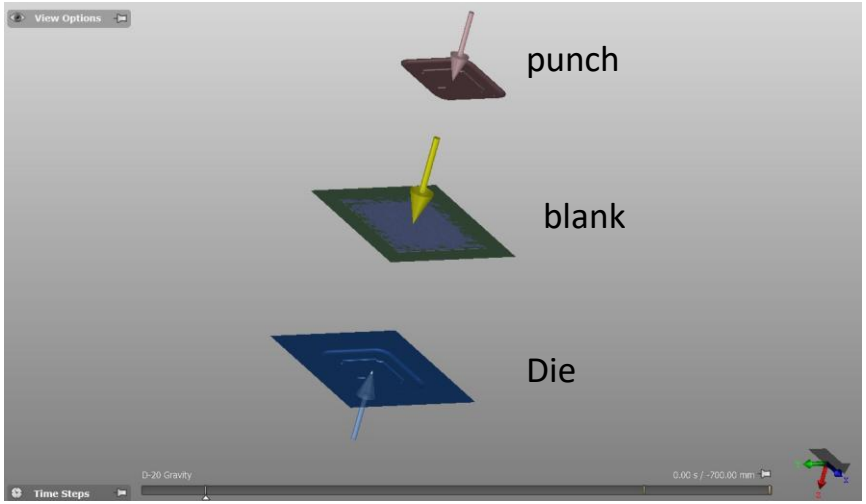
Punch, Die & Binder defined as rigid surfaces

Friction

Coulomb model; μ ≈ 0.10 – 0.15 (lubricated)

Binder Force

130 kN distributed over the binder area

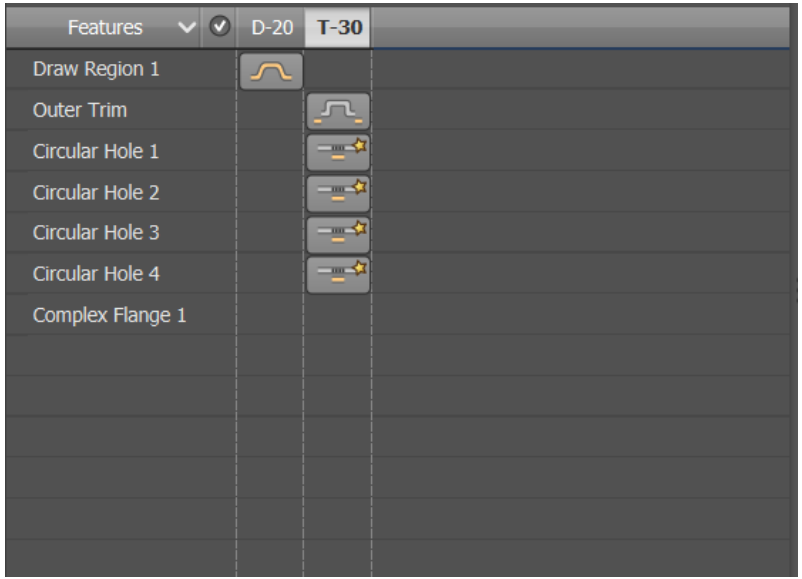


4. Process Setup

Operation type and forming stages

Operation Type

- Deep Drawing / Stamping (single-action)
- Outer Trimming
- Hole trimming

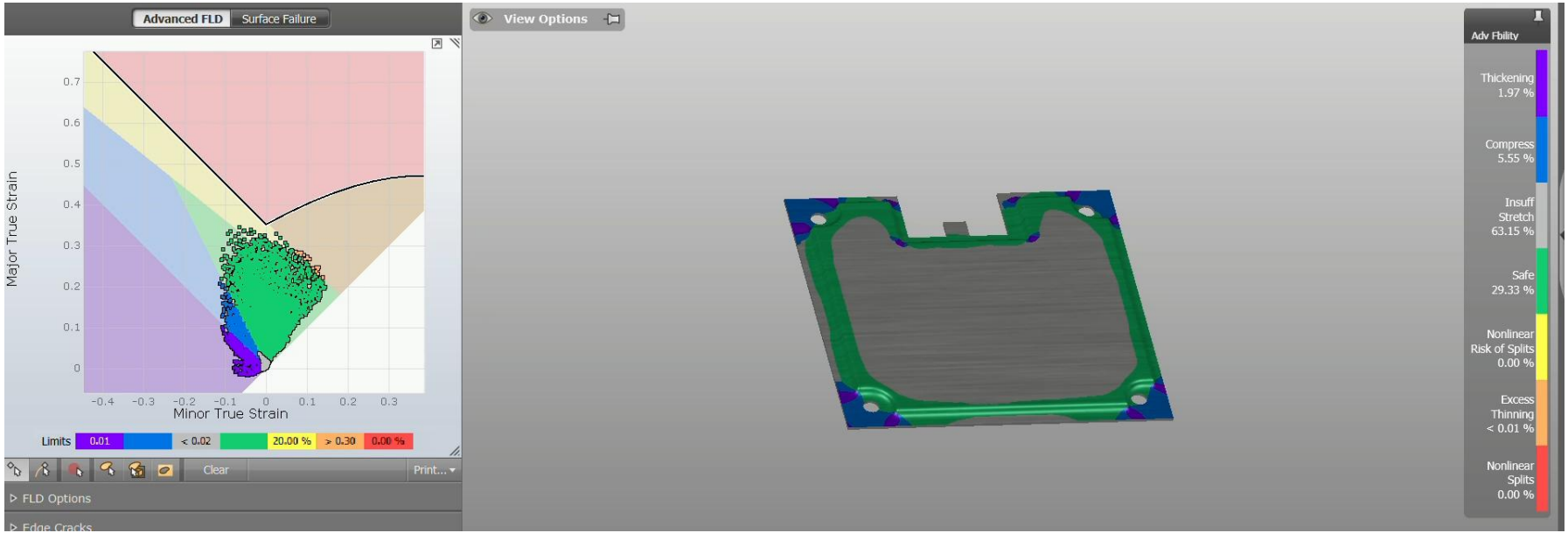


Forming Stages



Sequence executed in AutoForm Solver

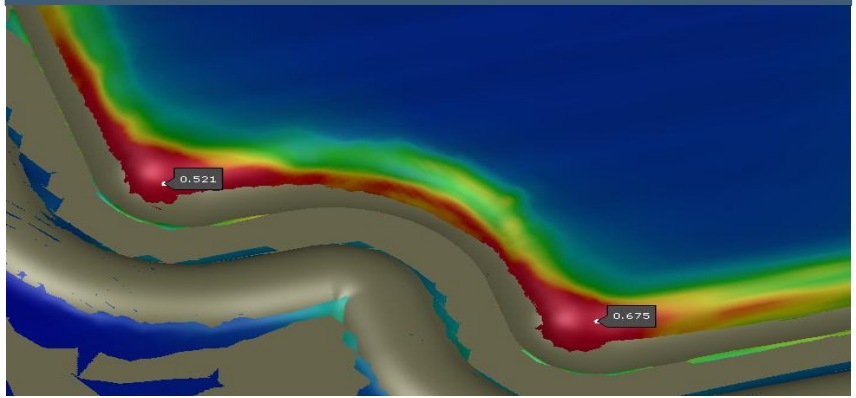
Forming Simulation



5. Iterations & Process Improvements

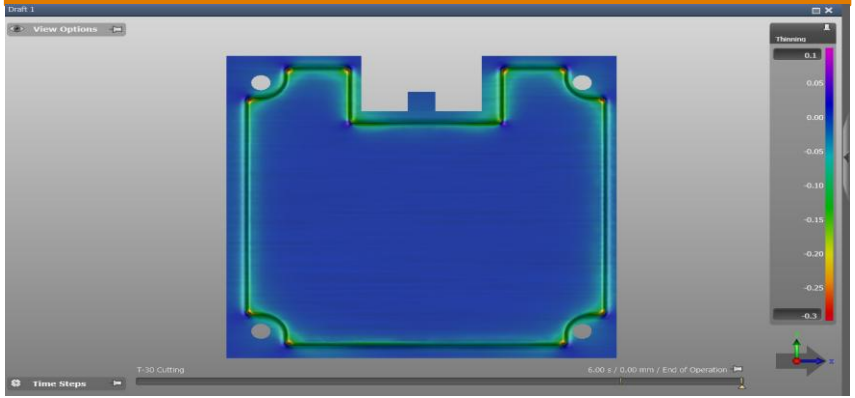
Initial defects → Modifications → Improved flow

Initial Iteration – Defects Observed



- Excessive thinning at corners
- Wrinkles in flange / blank holder area
- Cracks predicted on FLD

After Modifications – Improved Flow

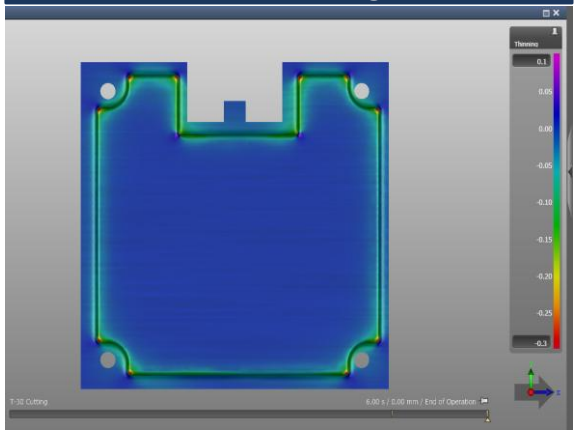


- Binder force re-tuned for uniform draw-in
- Fillet radius increased
- Friction coefficient reduced

6. Results

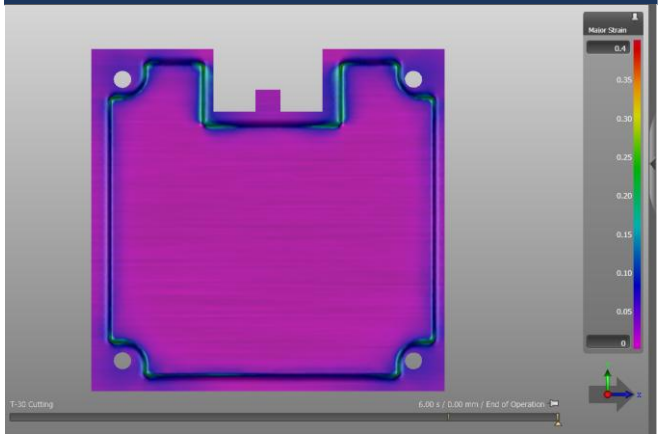
Thickness, strain & defect distribution

Thickness / Thinning



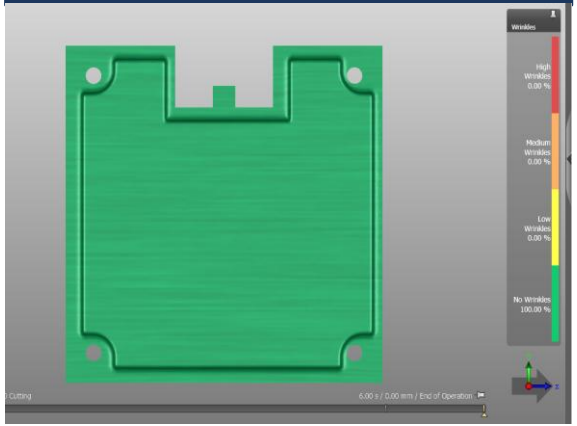
Min thickness: 0.3
Max thinning: 0.1

Strain Distribution



Major strain
Maximum: 0.4

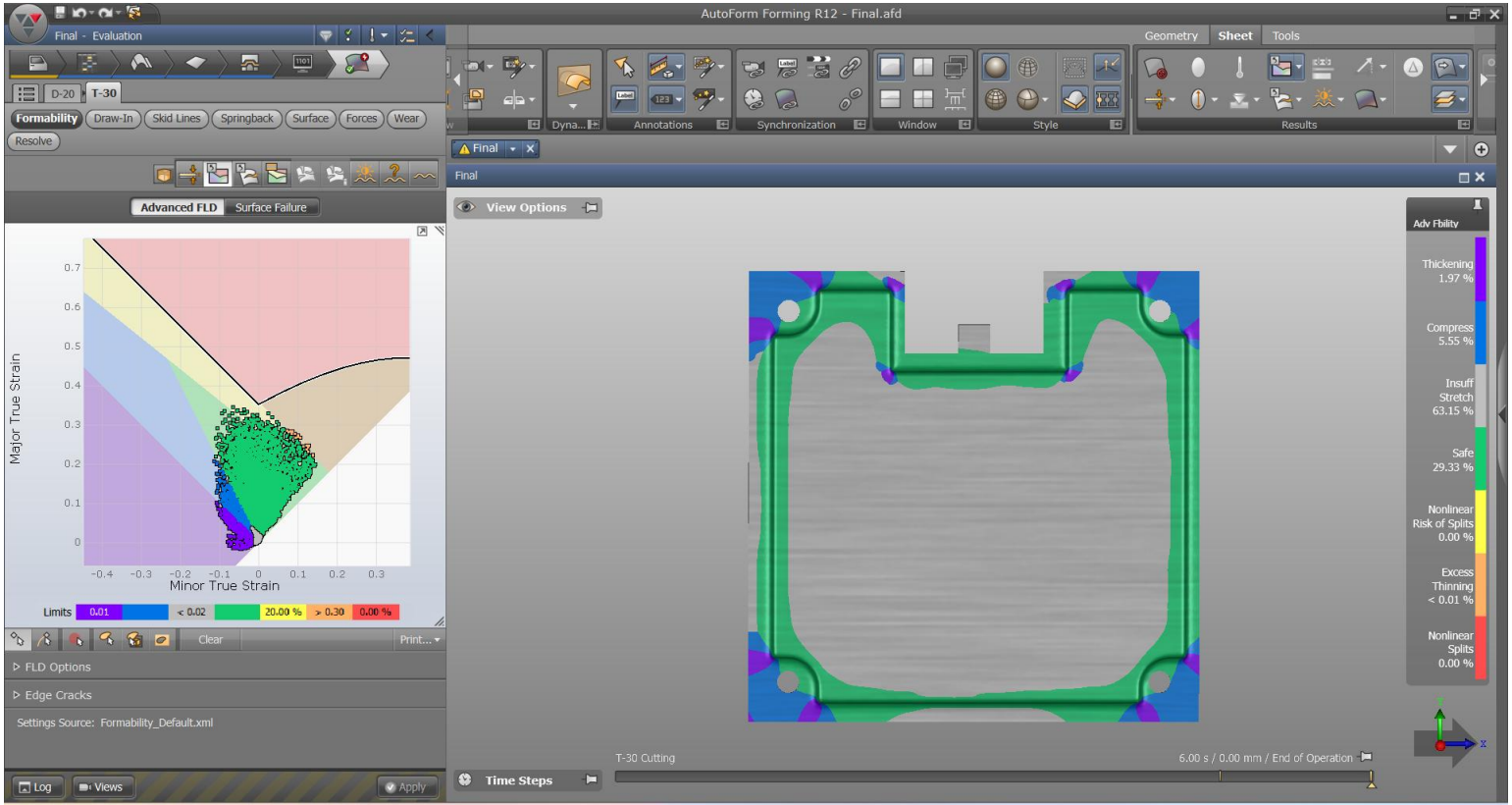
Wrinkles & Cracks



Cracks/Wrinkling: none

7. Failure Check

FLD, thinning limit, formability margin & process window



8. Forces

Press tonnage

Press Tonnage Required

Peak punch force ≈ 121.5kN (≈ 12.39tons)



9. Conclusion

Process stability and production feasibility

01

Stable Process

Defect-free part with thinning and strain within safe limits

02

Production Feasibility

Press tonnage & material flow acceptable for production

03

Optimised Parameters

Binder force, Friction coefficient, and geometry tuned for a robust process window

Thank You